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Organizational agility unleashed: Tapping intellectual capital for digital transformation effectiveness

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ABSTRACT

Digital transformation has become a critical and inevitable priority for organizations navigating increasingly dynamic and competitive environments. However, the mechanisms through which intellectual capital drives digital transformation effectiveness remain underexplored. This study examines the mediating role of organizational agility in this relationship, highlighting how its three dimensions—workforce, operational, and network agility—enable organizations to adapt and thrive in the digital era. A quantitative, correlational, and cross-sectional survey design was employed, underpinned by a post-positivist approach. Data were collected through a multi-stage sampling method from 265 respondents across 10 organizations in China and Thailand, representing various industries. The hypothesized mediation model was tested using PLS-SEM. The findings reveal that organizational agility fully mediates the effect of intellectual capital—comprising human, structural, and relational capital—on digital transformation effectiveness. Each dimension of organizational agility plays a distinct yet complementary role in enhancing digital transformation outcomes. This study uniquely contributes to the literature by providing a quantitative analysis of organizational agility's mediating role across its three dimensions. It advances the understanding of organizational agility as a multi-dimensional higher-level dynamic capability, bridging critical gaps in dynamic capabilities and digital transformation research. By elucidating these mechanisms, the study provides practical insights for organizations aiming to leverage intellectual capital and agility to drive transformative success, navigate uncertainty, and sustain a competitive advantage.

1. Introduction

Digital transformation goes beyond a mere technological shift (Henriette et al., 2015); it represents a complex, organization-wide transformation process that integrates technology with human and organizational factors, such as leadership, talent development, strategy, organizational culture, and strategic alliances (Fatima and Masood, 2024; Gong and Ribiere, 2021; Goran et al., 2017; Kane et al., 2017; Nadkarni and Prügl, 2021; Westerman et al., 2014). However, effective digital transformation requires more than just technological investment—organizations must also leverage their intangible assets, particularly intellectual capital (IC), to navigate uncertainty and sustain competitive advantage. IC is a critical intangible asset that encompasses accumulated knowledge, expertise, and relational networks that foster the knowledge and capabilities needed to drive digital transformation (Nwankpa et al., 2022; Subramaniam et al., 2019). While knowledge management focuses on how knowledge is created, shared, and used to create value, IC emphasizes the intangible knowledge assets that exist within an organization (Dabić et al., 2021; Paoloni et al., 2020). Yet, how can organizations effectively mobilize IC into successful digital transformation outcomes?

Some research has investigated IC's role in fostering innovation, new product development, innovative performance, organizational learning, and dynamic capabilities (Afshari and Hadian Nasab, 2021; Costa et al., 2014; Dinu, 2025; Oliveira et al., 2020; Thanh Nhon et al., 2020; Tsakalerou, 2015), while others have conceptually highlighted organizational agility (OA) as a strategic enabler of DT (e.g., Vial, 2019; Warner and Wäger, 2019). This study address the above mentioned question by examining the role of OA as a mediating mechanism between IC and digital transformation effectiveness (DTE). DTE refers to the extent to which an organization successfully achieves its intended strategic and operational digital transformation objectives, such as business model innovation, new value creation, customer engagement, and stakeholder satisfaction, etc. (Bresciani et al., 2021; Kane, 2019; Vaska et al., 2021).

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Yet, DTE is often context-dependent—organizations may perceive digital transformation success differently based on their goals, priorities, and capabilities. Drawing on the dynamic capabilities framework (Teece, 2018), this study conceptualizes OA as a higher-order dynamic capability that enables organizations to reconfigure and deploy IC in response to rapidly changing digital environments. OA reflects an organization's capabilities to sense digital shifts, seize opportunities, and transform resources and knowledge rapidly (Teece, 2018; Warner and Wäger, 2019). Prior research suggests that OA is essential for managing DT processes, as it enables organizations to integrate new technologies while responding to shifting market demands (Chakravarty et al., 2013; Felipe et al., 2016; Warner and Wäger, 2019). However, the mediating role of OA in the relationship between IC and DTE has not been sufficiently explored in empirical research. Limited empirical evidence exists on how IC translates into DTE through the mechanism of OA.

This study seeks to fill this gap by empirically examining the mediating role of OA in the IC-DTE relationship. OA is recognized as the core mechanism for building dynamic capabilities and organizational strategic renewal (Warner and Wäger, 2019), yet how it helps unlock and optimize IC to drive DT receives insufficient attention. We propose that IC enhances OA by providing organizations with the knowledge assets needed to detect market changes, make adaptive decisions, and implement strategic actions (Chung et al., 2012; Du Plessis, 2005; Jabeen et al., 2023; Pitafi et al., 2023). In turn, OA facilitates the execution of DT initiatives by reducing resistance to change and improving flexibility in DT implementation (Altschuller et al., 2010). Despite some scholars suggest that DT can bring changes that facilitate IC within an organization (Le et al., 2024; Ngo et al., 2023), this study focuses on the forward relationship—how IC fosters DT success through OA—based on the premise that knowledge assets are foundational to adaptive organizational behavior. In this way, we build on the dynamic capabilities theory to propose that IC is a critical input, OA is the mediating dynamic capability, and DTE is the outcome. By empirically testing this mediated relationship, this study contributes to a more nuanced understanding of the complex interactions between organizational resources and capabilities that drive DT success. In light of these consideration, we investigate the following research question in this paper:

To what extent does intellectual capital impact digital transformation effectiveness with the mediating effect of organizational agility?

To address our research question, we first establish a clear theoretical foundation grounded in existing literature on intellectual capital (IC), organizational agility (OA), and digital transformation effectiveness (DTE). We then propose a conceptual framework and related hypotheses examine the relationships among these constructs. We then outline the methodology and research design, including data collection and analysis. Finally, we discuss our findings, offering insights into theoretical contributions and practical implications.

2. Literature review

2.1. Digital transformation: a capability-based view

Digital transformation (DT) has become an imperative for organizations seeking to remain competitive and sustainable in rapidly changing environment. Organizations have to deal with it in a holistic and integrated way (Broccardo et al., 2024). Defined as a fundamental, organization-wide strategic transformation, DT requires continuous adaptation of knowledge, capabilities, structures, and processes to harness digital technologies effectively (Nadkarni and Prügl, 2020).

To understand how organizations navigate DT, this research draws on dynamic capabilities theory (Teece et al., 1997), which posits that organizations must integrate, build, and reconfigure internal competencies to address, or in some cases to bring about, changes in the business environment (Teece, 2007). Intellectual capital (IC)—a key intangible asset encompassing human, structural, and relational knowledge assets that drive innovation and strategic renewal (Edvinsson and Malone, 1997). Note that IC is the stock of knowledge embedded within an organization, while knowledge refers to processes of creation, sharing, and utilization (Dabić et al., 2021).

Dynamic capabilities theory provides a useful lens to explain how IC enables organizations to sense opportunities, seize technological advancements, and transform organizational structures to achieve digital transformation success (Gong and Ribiere, 2025; Li et al., 2018). However, the full potential of IC remains unrealized without an intermediary mechanism that facilitates the translation of knowledge into adaptive strategic actions. We propose that organizational agility (OA) acts as this key mechanism, linking IC to DT effectiveness by fostering flexibility and rapid decision-making (Felipe et al., 2016).

2.2. Intellectual capital as a strategic knowledge resource

The relevance of IC as a vital intangible asset for organizations started to emerge around the 1980s, primarily among practitioners, and later gained traction in academic field around 1994 (Dabić et al., 2021). Since then, IC research has evolved through four major stages within the management domain (Dumay and Garanina, 2013; Guthrie and Petty, 2000; Lin and Edvinsson, 2020; Mouritsen, 2006; Serenko and Bontis, 2013). The concept of IC encompasses "a combination of intellect and capital to convey the importance of knowledge" (Serenko and Bontis, 2013, p. 478). It originates from knowledge creation processes and the development of new relationships (Konno and Schillaci, 2021); yet, it is importance to distinguish IC from knowledge management. While IC refers to knowledge assets embedded within the organization, knowledge management focuses on how knowledge is created, shared, and used to create value (Paoloni et al., 2020).

A widely accepted conceptualization of IC include its tripartite representation as human capital, structural capital, and relational capital (Dabić et al., 2021; Guthrie et al., 2012). Human capital (HC) refers to the sum of the skills, experience, capabilities, and innate knowledge of workers (Edvinsson and Malone, 1997); the knowledge embedded in people (Guthrie et al., 2012). It represents the human factor (i.e., the combined intelligence, skills, and expertise) that gives the organization its distinctive character (Bontis et al., 2002). HC serves as the inventive mechanism for creativity, innovation, and strategic renewal. It significantly influences an organization's ability to develop and deploy digital solutions effectively (Afshari and Hadian Nasab, 2021). Structural capital (SC) refers to the knowledge an organization has been able to internalize and that remains in the organization, be it a structural, procedural, or cultural element when the employees leave the organization (Bontis et al., 2000; Petrash, 1996). It is the knowledge embedded in the organization and its systems (Guthrie et al., 2012) that encourages the human resource to create and leverage its knowledge (Edvinsson and Sullivan, 1996). Its main goal is to support the conversion of HC into IC. Hence, it was described as the backbone of the organization (Burr and Girardi, 2002), the "institutionalized knowledge" (De Pablos, 2004, p. 629), or the supportive infrastructure of HC embedded in organizations (Bontis, 1999; Chen, 2009). It plays a pivotal role in systematizing digital initiatives and fostering knowledge integration across the organization (Kudyba, 2020). Relational capital (RC) refers to all knowledge assets acquired through the organization's direct and indirect relationships and interactions with various external stakeholders, including customers, suppliers, cooperation partners, investors, rivals, government, and others, i.e., all external stakeholders (Martini et al., 2016). Sveiby (2000) calls this dimension an external component of IC (Gibbert et al., 2001). It is the sum of all assets that arrange and manage the organization's relations with the environment (Bozbura, 2004; Bozbura et al., 2007; Lee, 2010; Martínez-Torres, 2006). This dimension is based on the idea that organizations cannot be perceived as isolated systems in today's business environment, but rather that they depend significantly on the relations they build up in their environment (Hormiga et al., 2011). RC is critical for enhancing the organization's capacity to respond to external changes and digital market dynamics (Fan and Liu, 2024).

Recent studies emphasize that organizations with strong IC are better positioned to succeed in DT by leveraging their knowledge assets to drive innovation and digital adoption (Dost et al., 2016; Duodu and Rowlinson, 2019; Inkinen, 2016). Each dimension of IC has been shown to contribute positively to DT outcomes (Pang et al., 2025). However, the mere possession of IC does not guarantee DT success. Organizations must possess the capability to translate IC into adaptive and strategic actions aligned with the evolving digital environment (Yeow et al., 2018). This underscores the need for an intermediary mechanism that enables organizations to dynamically leverage and reconfigure IC in response to digital challenges and opportunities—a role we argue is fulfilled by organizational agility.

2.3. Organizational agility as a dynamic capability

The term “agility” was initially introduced in 1982 within the business context as “corporate agility,” referring to “the capacity to react quickly to rapidly changing circumstances, requires a focus on clear system output goals and the capability to match human resources to the demands of changing circumstance” (Brown and Agnew, 1982, p. 29). However, it was not until the 1990s that the concept gained broader recognition among scholars and industry professionals. Agile manufacturing emerged during this period as a customer-focused manufacturing strategy and later evolved—particularly after 2000—to encompass organizational and process-oriented perspectives (Wendler, 2013). It gained significant prominence in software engineering in the early 2000s (Fowler and Highsmith, 2001). With the rise of service-product systems and Internet-dominant businesses, agility has been extended to non-manufacturing organizations, fostering literature around enterprise and organizational agility (Nejatian et al., 2019; Shin et al., 2015). In today’s volatile and competitive environment, organizations often face a trade-off between flexibility and cost efficiency. This makes it crucial to develop cost-effective yet high-performing agile capabilities (Akter et al., 2023). Agility is understood as a dynamic capability (Talaei-Khoei et al., 2024)—a context-specific, change-embracing, and growth-oriented approach (Goldman et al., 1995). It enables organizations to thrive in fragmented and fast-paced global markets, thereby improving overall performance (Dikert et al., 2016). Importantly, organizational agility (OA) extends beyond mere responsiveness or speed, necessitating fundamental structural and infrastructural changes for long-term survival in a VUCA (Volatility, Uncertainty, Complexity, and Ambiguity) world (Oliva and Kotabe, 2019; Youssef, 1994).

OA is both reactive—allowing quick adaptation—and proactive—enabling organizations to anticipate and capitalize on emerging opportunities (Alavi et al., 2014). Scholars emphasize proactivity as a defining characteristic of OA, encompassing deliberate actions that shape, rather than merely respond to, environmental change (Dove, 2002; Doz and Kosonen, 2007; Sharifi and Zhang, 1999; Sherehiy et al., 2007). Therefore, considering the dual nature of OA, we define it as an organization’s ability to quickly respond and proactively embrace unanticipated changes in dynamic environments. Such an ability should be reflected in the core aspects of an organization (i.e., workforce, operation, and network); thus, deriving different dimensions of OA. Based on IC theory, Gong and Ribiere (2025) conceptualized OA as a higher-order dynamic (learning) capability comprised of three lower-order dynamic (functional) capabilities: workforce agility, operational agility, and network agility. These sub-dimensions reflect an organization’s ability to reconfigure resources effectively and swiftly in support of DT.

Strategic responsiveness in managing IC is essential for nurturing agility (Singhania and Panda, 2025). Organizations with strong IC exhibit greater workforce agility to enhance their ability to analyze, predict, and act on digital opportunities and emerging technologies (Bresciani et al., 2022; Li et al., 2021). Structural capital plays a critical role in fostering operational agility by providing organizational routines, processes, and mechanisms for knowledge transfer, flexible workflows, and rapid implementation (Chakravarty et al., 2013). Relational capital supports network agility by facilitating collaboration, trust, and real-time communication with external stakeholders, which are vital for navigating digital ecosystems (Greer et al., 2017; Zhao et al., 2023). However, despite these connections, the existing literature often treats IC and OA as separate constructs, and few empirical studies have examined their interaction in the context of DT. While some scholars argue that DT itself contributes to the development of IC (Ngo et al., 2023), we posit the reverse: organizations with strong IC are better equipped to develop OA, which then enhances DT effectiveness. This perspective aligns with dynamic capabilities theory, which emphasizes that knowledge assets must be continuously reconfigured and leveraged to adapt to digital environments (Teece, 2018). OA, as a mediating mechanism, enables the dynamic mobilization of IC to meet the demands of DT.

2.4. Digital transformation and digital transformation effectiveness

DT radically modifies an organization’s business models, strategic orientation, and knowledge, posing significant challenges for organizations worldwide (D’Ippolito et al., 2019; Nambisan et al., 2019; Ståhle et al., 2015; Švarc et al., 2020; Warner and Wäger, 2019). The success of DT extends beyond the timely and budget-compliant delivery of technological artifacts (Nwankpa, 2015); it also hinges on people dimensions, including employee attitude, perceptions, and ownership of DT benefits (Badewi, 2022). As such, DT is a strategically enabled transformation that depends on the effective orchestration of key organizational resources and capabilities (Gong and Ribiere, 2021). In the organizational context, Gong et al. (2023) defined DT as “a fundamental change process of an organization enabled by exploring the use of digital technologies to redefine its business models” (p. 303). This process is inherently complex, typically comprising multiple concurrent initiatives—technological, human, and cultural—that are often structured and executed as distinct projects (Vial, 2019; Warner and Wäger, 2019). Given this project-based nature of DT, our study focuses on evaluating the success of DT projects as a proxy for overall DT effectiveness at the organizational level.

According to the PMBOK© Guide, a project is defined as “a temporary endeavor undertaken to create a unique product, service, or result.” In project management discourse, the distinction between project efficiency (doing things right) and project effectiveness (doing the right things) is critical, although often blurred in literature and practice (Belout, 1998; Ika, 2009). Drawing on Zidane and Olsson’s (2017) systematic literature review, project efficiency is more about comparing the outputs of the project to its inputs, whereas project effectiveness “occurs once the operation of the produced product [and service] generates positive impacts in the middle and long term” (p. 634). Following their definitions, we conceptualize digital transformation effectiveness (DTE) as the extent to which the digital transformation project objectives are achieved regarding the results generated. In other words, DTE refers to the degree to which an organization’s digital initiatives fulfill their intended strategic and operational objectives. Importantly, DTE is not a one-size-fits-all construct; different organizations may define effectiveness based on criteria such as increased efficiency, enhanced customer experiences, improved employee engagement, new value creation, or new business models (Kane et al., 2017).

3. Research framework and hypotheses development

3.1. Intellectual capital and organizational agility

IC involves knowledge needed for an organization to deal with the current business environment characterized by growing complexity and dynamicity, rapid changes, and uncertainty, where OA is vital and offers the basis for survival or a competitive advantage. OA reflects an organization's ability to sense changes in the environment and swiftly respond through strategic reconfiguration of resources and competencies. The relationship between IC and OA can be grounded in dynamic capability theory (Sandeep et al., 2025), which posits that an organization's ability to reconfigure internal and external competencies to address the rapidly changing environments is driven by its intangible assets. Under this theoretical lens, IC acts as a fundamental driver that enables organizations to develop the OA needed to navigate volatile market conditions (Ahmed et al., 2022). As intangible assets, the components of IC serve as the knowledge infrastructure that facilitates learning, adaptation, and innovation, which are essential for OA. Prior research underscores the role of effective knowledge sharing and transfer in fostering OA (Chung et al., 2012; Du Plessis, 2005). Organizations must leverage both internal and external knowledge to identify opportunities and threats in the market and align them with strategic objectives (Cegarra-Navarro and Martelo-Landroguez, 2020). The process of knowledge creation and utilization, as highlighted by Altschuller et al. (2010) enhances the development of OA by providing organizations with the necessary cognitive and procedural resources to swiftly adapt. Additionally, empirical research by Maditinos et al. (2014), Nafei (2016), and Nissen and von Rennenkampff (2017) demonstrate that organizations with strong IC exhibit higher levels of OA as the ability to manage and apply knowledge effectively in response to market dynamics and industry disruptions. That is, OA focuses on managing the knowledge mentioned above effectively to respond to and deal with changes.

Breaking intellectual capital (IC) into its three core dimensions helps further elucidates its role in enhancing organizational agility (OA). Human capital, comprising employees' knowledge, strengthens workforce agility (WA) by fostering continuous learning, cross-functional collaboration, and innovation. Employees with high levels of expertise, creativity, and adaptability enable rapid innovation and problem-solving. The cognitive diversity and absorptive capacity of employees allow organizations to sense and interpret changes in the environment effectively. More importantly, learning-oriented employees facilitate continuous renewal of capabilities, a core requirement for OA (Teece et al., 2016). Structural capital, which represents embedded organizational knowledge in databases, processes, and culture, enhances operational agility (OpA) by enabling efficient knowledge transfer, streamlined decision-making, and flexible business processes (Ahn et al., 2018). Flexible structures, decentralized decision-making processes, and efficient knowledge management systems reduce internal friction and speed up execution. A knowledge-sharing culture and supportive IT systems facilitate fast internal coordination and organizational learning. Relational capital, which refers to an organization's external partnerships and customer relationships, facilitates network agility (NA) by ensuring the rapid exchange of market intelligence and fostering adaptive collaborations. External collaboration networks enable join problem-solving and co-creation of new solutions when facing sudden environmental changes or under turbulent conditions (Ahmed et al., 2022). The three dimensions of IC serve as enablers for the different facets of OA because they provide the foundation for acquiring, transferring, and applying knowledge in ways that enhance the organization's ability to respond to change. Empirical evidence supports the direct impact of IC on OA across industries. Ghafuri et al. (2014) demonstrated that IC positively influences OA within the service sector, while Baharloo (2016) and Al-Azzam et al. (2017) found a strong positive correlation between IC dimensions and OA in educational and business settings, respectively. Additionally, Shami and Nastiezaie (2017) highlighted that IC enhances OA through organizational learning mechanisms. Based on the theoretical justifications and empirical findings discussed above, we propose the following hypotheses:

H1. There is a positive relationship between intellectual capital (IC) and organizational agility (OA).

H1a. There is a positive relationship between human capital (HC) and workforce agility (WA).

H1b. There is a positive relationship between structural capital (SC) and operational agility (OpA).

H1c. There is a positive relationship between relational capital (RC) and network agility (NA).

3.2. Organizational agility and digital transformation effectiveness

OA plays a pivotal role in managing the uncertainties and risks associated with DT, which involves multifaceted changes at technological, structural, and cultural levels (Khoramgah, 2012). Gong and Ribiere (2025) reinforced this idea through their integrative literature review and emphasized that OA enables organizations to navigate the volatility of digital initiatives by fostering adaptability and resilience. In today's dynamic and highly competitive environments, OA has transitioned from a strategic advantage to an essential requirement for organizational survival and success (Felipe et al., 2016; Sambamurthy et al., 2003). Grounded in dynamic capabilities theory, OA serves as a key enabler of DT effectiveness. However, while OA enhances an organization's ability to integrate new technologies more fluidly and adjust to changing digital landscapes, it does not inherently guarantee successful transformation outcomes. The effectiveness of DT is contingent upon multiple contextual factors, such as the suitability of digital technologies, their alignment with organizational strategy, and the organization's ability to address customer concerns while mitigating unintended consequences.

However, agility within an organization's workforce, operations, and networks facilitates the rapid exchange of ideas and experiences among its members, fostering an environment that promotes risk-taking and experimentation during the implementation of DT projects. Workforce agility enhances an organization's capacity for rapid learning, strategic sensing, and quick adaptation to emerging digital tools, platforms, and workflows. Operational agility ensures that cross-functional teams to rapidly reallocate resources and adjust priorities in response to evolving DT project demands. Network agility enables organizations to leverage external expertise, foster open innovation, and enhance collaboration with ecosystem partners, allowing DT projects to align more effectively with market shifts. These three dimensions of OA are essential for navigating the complexities and uncertainties inherent in DT projects, leading to more effective project outcomes.

By fostering real-time responsiveness and flexibility, OA enables organizations to adjust internal structures and decision-making processes (Tsourveloudis and Valavanis, 2002), thereby increasing the likelihood of achieving DT objectives. Moreover, OA expands an organization's capacity for innovation—whether through business model innovation or radical technological shifts—by facilitating rapid adaptation and execution (Chakravarty et al., 2013). This OA-driven transformation is critical for flexible operations, sustaining competitive advantage in the digital economy (Lu and Ramamurthy, 2011). In this context, capability-driven outcomes such as business model innovation, radical changes in offerings, strategic reconfiguration of digital technologies become integral to successful DT (Gong and Ribiere, 2021). Based on this discussion, the following hypotheses are proposed:

H2. There is a positive relationship between organizational agility (OA) and digital transformation effectiveness (DTE).

H2a. There is a positive relationship between workforce agility (WA) and digital transformation effectiveness (DTE).

H2b. There is a positive relationship between operational agility (OpA) and digital transformation effectiveness (DTE).

H2c. There is a positive relationship between network agility (NA) and digital transformation effectiveness (DTE).

3.3. The mediating effects of organizational agility

DT research has evolved beyond a technology-centric focus to emphasize the human, structural, and relational dimensions that enable organizations to navigate complex and uncertain environments. Successful DT requires more than just technological advancements; it demands alignment across various organizational elements, particularly those rooted in IC. IC provides the knowledge foundation for DT by fostering human-driven innovation (i.e., HC dimension), adapting internal processes and structures (i.e., SC dimension), and leveraging external knowledge and stakeholder cooperation (i.e., RC dimension). These dimensions enable organizations to absorb, process, and apply new knowledge in ways that enhance DTE (Dremel et al., 2017; Vial, 2019). Given the intangible nature of IC, its impact on DTE is not always direct but operates through intermediary mechanisms that enhance adaptability and responsiveness. While IC equips organizations with the necessary knowledge base, OA serves as the critical mechanism through which this knowledge is operationalized. OA enables organizations to reconfigure resources, sense opportunities, and respond to digital disruptions in real-time (Gong and Ribiere, 2025). In environments characterized by rapid technological shifts, organizations must rely on OA to maximize the value of IC, ensuring that knowledge assets translate into actionable strategies that drive effective DT.

We argue that OA plays a pivotal mediating role in translating IC into effective DT by facilitating knowledge application and transfer embedded in IC into tangible outcomes for DT projects. Specifically, OA ensures that organizations can proactively manage changes at multiple levels and adjust internal structures to align with emerging market needs (Van Oosterhout et al., 2006), enabling organizations to quickly adapt to specific demands of each local context (Piening et al., 2016; Schneckenberg, 2015). OA supports the rapid search and retrieval of relevant knowledge, allowing organizations to leverage it in developing high-quality offerings or in responding to the emergence of new rivals (Cegarra-Navarro et al., 2016). This OA-driven adaptability enables organizations to optimize processes, allocate resources efficiently, and enhance decision-making, ultimately leading to more effective DT initiatives (Tallon and Pinsonneault, 2011). Organizations that fail to develop OA despite processing strong IC may encounter rigid decision-making structures, delayed responses to change, and resistance to digital integration, ultimately undermining the success of DT initiatives. Thus, building on theoretical and empirical gaps, this study proposes that IC contributes to DT effectiveness indirectly through the development of OA.

Each dimension of OA plays a unique mediating role in translating IC into DTE: Workforce agility enhances human capital by ensuring that employees can rapidly acquire, apply, and adapt knowledge to evolving business challenges. Employees' capability to experiment, learn, and adjust in response to digital shifts is crucial for DTE (Dremel et al., 2017; Vial, 2019). Workforce agility ensures that human capital remains dynamic and responsive rather than static, fostering continuous learning and innovation throughout DT initiatives (Lu and Ramamurthy, 2011). Operational agility supports structural capital by facilitating rapid adjustments in workflows, systems, and operational processes to accommodate new technologies seamlessly. Structural capital provides the foundation for DT, but without operational agility, organizations may struggle to realign operations with digital advancements. OpA allows organizations to reconfigure internal systems and processes efficiently, ensuring that DT projects can adapt in real-time to technological and market demands. Network agility amplifies relational capital by enhancing external knowledge flows and collaboration, enabling organizations to co-create digital solutions, integrate external expertise, and refine product and service portfolios (Chesbrough et al., 2006; Trinh et al., 2012). Organizations that develop strong relational capital but lack network agility may fail to capitalize on these relationships for DT. Network agility ensures that organizations can quickly leverage external insights and collaborations, enhancing their ability to align DT initiatives with evolving market expectations.

By fostering internal (HC), structural (SC), and external (RC) knowledge flows, OA serves as a bridge between IC and DTE. It enables organizations to integrate internal expertise, structural flexibility, and external collaborations, ensuring that knowledge assets are fully leveraged for digital transformation. Organizations that fail to develop OA will struggle to adjust processes and routines in response to the signals that IC provides for DT (Cegarra-Navarro et al., 2016), leading to stagnation in transformation efforts. Conversely, organizations that cultivate high levels of OA will effectively align their knowledge assets with digital demands (Alegre and Sard, 2015; Shahrabi, 2012), increasing their chances of achieving DTE. Based on the discussion above, we propose the following hypotheses:

H3. Organizational agility (OA) mediates the relationship between intellectual capital (IC) and digital transformation effectiveness (DTE).

H3a. Workforce agility (WA) mediates the relationship between human capital (HC) and digital transformation effectiveness (DTE).

H3b. Operational agility (OpA) mediates the relationship between structural capital (SC) and digital transformation effectiveness (DTE).

H3c. Network agility (NA) mediates the relationship between relational capital (RC) and digital transformation effectiveness (DTE).

Overall, this research attempts to explore the extent to which the three dimensions of IC impact DTE, considering the mediating role of OA (as shown in Fig. 1). This study adopts dynamic capabilities theory as the guiding theoretical lens to explain how organizations mobilize internal intangible assets to adapt to digital transformation. Specifically, we conceptualize intellectual capital (IC) as a foundational knowledge resource that enables organizations to build higher-order capabilities. Organizational agility is positioned as a key dynamic capability that mediates the relationship between IC and digital transformation effectiveness (DTE).

4. Methodology

This study adopted a post-positivist worldview and a quantitative, non-experimental, correlational, and cross-sectional survey research design to examine the proposed relationships. This approach was chosen to establish measurable relationships between variables, aligning with the study's objectives and the nature of the variables involved. Post-positivism facilitates hypothesis testing while acknowledging researcher bias and limits of generalizability. Given the complexity of DT, this research focuses on DT projects that fall under DT's broader umbrella, but are framed around discrete DT efforts in the areas of employee engagement, operational efficiency, data management, customer engagement, and new value creation. These projects, although operationally narrower, serve as pivotal building blocks for long-term strategic DT initiatives that reshape an entire organization. This positioning aligns with the research's emphasis on organizational agility as an underlying mechanism that enables both short-term and cumulative DT impacts. Hence, the target population consisted of managers and employees involved in completed DT projects across organizations. A multi-stage sampling method combining non-probability and probability techniques was employed. In the first stage, critical case purposive sampling, following Saunders et al.'s (2019) guideline, identified ten organizations of varying sizes and ages to enable comparative insights. In the second stage, stratified sampling was used to select participants from three hierarchical strata—senior executives, middle managers, and project team members—ensuring balanced representation. This sampling design allowed the study to address its research questions effectively while ensuring diverse perspectives from all levels of organizational involvement in DT projects.

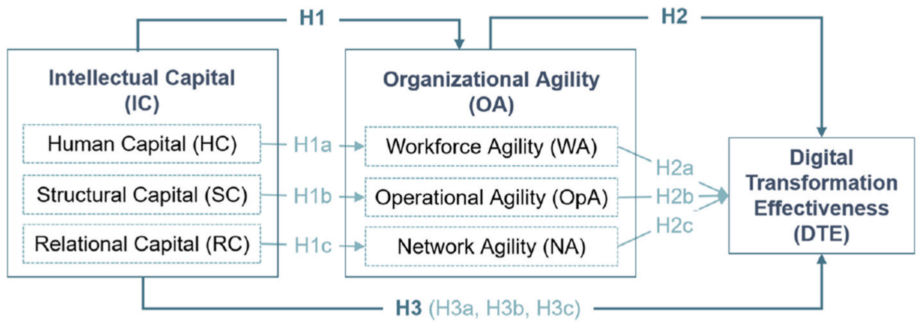

Fig. 1. The proposed conceptual framework.

4.1. Measures and instruments

The research questionnaire consists of four parts: measures for Intellectual capital (IC: HC, SC, RC), Organizational agility (OA: WA, OpA, NA), digital transformation effectiveness (DTE) through perceived DT project effectiveness, and participant descriptive data. The first three sections employ perception-based questions to assess IC, OA, and DTE, drawing on participants' organizational experiences. These questions are adapted from previously validated survey instruments developed by Felipe et al. (2020); Kane (2019); Zhang et al. (2018); Doz and Kosonen (2010); Engelman et al. (2017); Lu and Ramamurthy (2011); Sharabati et al. (2010); Doz and Kosonen (2010), etc. (See Appendix). A multilevel model was employed to address the complexity of real-world scenarios, examining the organizational-level factors (IC, OA) and project-level outcomes (DTE). This approach aligns with the variables' nature and provide more accurate insights into how higher-level factors influence project outcomes. It also explains why the PLS-SEM Hierarchical Component Model is used to analyze proposed relationships.

To keep the scale's consistency, most items were measured using a five-point Likert scale (1 = Strongly agree to 5 = Strongly disagree), with exceptions for DTE1 (1 = Not close at all to 5 = Very close) and DTE7 (1 = Very unsuccessful to 5 = Very successful). The questionnaire items are included in the Appendix.

To ensure accuracy across languages, English and Thai native speakers performed translations, using a back-translation technique to compare and refine the Thai version. The same process was applied to create the Chinese version, ensuring consistency and equivalence between the source and target questionnaires.

4.2. Data collection and analysis

Two approaches were utilized for data collection: an online survey and workshops. Online surveys, while efficient for reaching participants, can be impersonal, leading to potential misinterpretation of questions. To address this limitation, four face-to-face workshops (2–2.5 h each) were held at the participating organizations' offices. Key DT stakeholders assessed their DT projects in small groups (4–5 people) using the same questions as the online survey. The dual approach ensured robust data collection, capturing numerical insights and contextual understanding directly from participants involved in DT projects.

Given the exploratory nature of this research, model complexity (e.g., two reflective-formative higher-order constructs and 7 constructs with 53 items), and a sample size of over 200, partial least squares (PLS) was used to test the structural equation models. A priori power analysis, based on Hair et al.'s (2021) update of Kock and Hadaya's (2018) inverse square root method, indicated a minimum sample size of 251 observations, as the construct DTE had six incoming paths in the model.

Data quality checks addressed issues such as missing data, suspicious response patterns, and outliers. Responses with low variance (standard deviation < 0.25) were removed, resulting in 265 valid responses from 333 collected across 10 organizations. Participants primarily worked in management, customer service, marketing, IT, and operations. Approximately half were managers, providing balanced input from leadership and team members. Most participants were aged 28–44, with over half employed at their organization for more than five years. Table 1 summarizes participant demographics.

5. Results

The proposed model was analyzed using the PLS-SEM Hierarchical Component Model (HCM) (Hair et al., 2017, 2021) and the disjoint two-stage approach (Agarwal and Karahanna, 2000; Becker et al., 2012), conducted in SmartPLS 3.2.9 (Ringle et al., 2015). In stage one, the measurement model for lower-order constructs was assessed using a standard model, establishing direct relationships. In stage two, latent variable scores from the lower-order constructs were used to estimate the higher-order model (Fig. 2).

PLS-SEM was chosen for its ability to handle complex models with multiple constructs, indicators, and smaller sample sizes. The HCM approach facilitated nuanced analysis of how various IC and OA components drive DTE while maintaining simplicity at the higher-order level. The disjoint two-stage approach ensured accurate and consistent estimation of both lower- and higher-order components. Incorporating higher-order constructs reduced collinearity among formative indicators by reorganizing them across concrete sub-dimensions of broader constructs (Sarstedt et al., 2019). This approach captured the overall effect of IC and OA on DTE while uncovering specific pathways through sub-dimensions. These techniques provided a deeper understanding of the mechanisms involved, aligning with the research objectives and enhancing the precision of the findings.

5.1. Measurement model assessment

Validating reflective lower-order constructs (LOCs). Construct reliability was established as both Cronbach's Alpha (ranged from 0.78 to 0.9) and composite reliability (CR, 0.85 to 0.92) exceeded the threshold of 0.7. Convergent validity was achieved, with AVE values surpassing 0.5 for all constructs: HC (0.52), SC (0.58), RC (0.51), WA (0.53), OpA (0.65), NA (0.62), and DTE (0.63). Discriminant validity was verified using the Heterotrait-Monotrait ratio (HTMT), adhering to the $HTMT_{0.9}$ criterion (Gold et al., 2001). To meet this standard, 7 items (e.g., HC7, HC9, WA1, WA3, OpA3, OpA7, OpA9) with cross-loadings below 0.1 were removed. Multicollinearitywas not an issue, as VIF values for all indicators remained below 5.

Validating reflective-formative higher-order constructs (HOCs). Latent variable scores from LOCs were used to estimate two HOCs, namely IC and OA. Content validity was ensured through a thorough literature review. As Sarstedt et al. (2013) suggested, convergent validity was assessed using a global reflective item for each HOC, measured on a five-point Likert scale. The IC global item is "in general, please

Table 1
Respondent-level and organization-level descriptive statistics.

Categories		Sub-Categories	Frequency	Percent
Participants	Functional Affiliation	General management	44	16.6 %
		Customer service	31	11.7 %
		Marketing	28	10.6 %
		Information technology	25	9.4 %
		Operations	17	6.4 %
		Sales	17	6.4 %
		Finance	11	4.2 %
		Product development	8	3 %
		Research	7	2.6 %
		Human resources	7	2.6 %
		Corporate communications	5	1.9 %
		Risk management	4	1.5 %
		Supply chain operations management	3	1.1 %
		Others	58	21.9 %
	Position level	C-suite executive	26	9.8 %
		Department manager	76	28.7 %
		Project director	24	9.1 %
		Technical staff	56	21.1 %
		Support staff	64	24.2 %
		Not indicated	19	7.2 %
	Working Years	Less than 2 years	52	19.6 %
		2–5 years	55	20.8 %
		5–10 years	64	24.2 %
		10+ years	74	27.9 %
		Not indicated	20	7.5 %
	Age	21 or younger	1	0.4 %
		22–27	39	14.7 %
		28–35	73	27.5 %
		36–44	57	21.5 %
		45–52	49	18.5 %
		53–59	26	9.8 %
		60 or older	1	0.4 %
		Not indicated	19	7.2 %
Organizations	Industry	Banking	52	19.62 %
		Automotive	43	16.23 %
		Marketing	41	15.47 %
		Insurance	24	9.06 %
		Manufacturing	21	7.92 %
		Financial (Investment)	20	7.55 %
		Software/Computers	20	7.55 %
		Telecommunications	20	7.55 %
		Agriculture	17	6.41 %
		Others	7	2.64 %
		Total	265	
	China	Chinese Company 1 (C1)	17	14.53 %
		Chinese Company 2 (C2)	22	18.8 %
		Chinese Company 3 (C3)	30	25.64 %
		Chinese Company 4 (C4)	24	20.51 %
		Chinese Company 5 (C5)	20	17.1 %
		Others	4	3.42 %
		Total	117	
	Thailand	Thai Company 1 (T1)	17	11.49 %
		Thai Company 2 (T2)	43	29.05 %
		Thai Company 3 (T3)	21	14.18 %
		Thai Company 4 (T4)	17	11.49 %
		Thai Company 5 (T5)	47	31.76 %
		Others	3	2.03 %
		Total	148	

assess the extent to which knowledge embedded in your people, your institutionalized knowledge, and knowledge accumulated from direct and indirect relationships have been utilized." That of OA is "in general, please assess the extent to which your workforce, operations, and networks can quickly respond and proactively embrace changes in dynamic environments." At the 95 % confidence interval, correlations between HOCs and their global measures met the threshold of 0.7. Collinearity was minimal, with VIF values for HOCs below 5. Outer loadings for LOCs were greater than 0.5 and significant, further validating the HOCs (Sarstedt et al., 2019). All validation criteria were satisfied, confirming the robustness of HOCs.

5.2. Structural model assessment

The structural model was evaluated by examining collnearity, significance of path coefficients, and model fit. All VIF values were below 5, indicating no critical collinearity issues among predictor construct. A 5000-sample bootstrapping routine was performed to compute *t* values and *p* values for structural path coefficients. The results are detailed in the "Mediation analysis" section. Model fit was assessed using the coefficient of determination (R^2), effect size (f^2), predictive relevance measure (Q^2), and the standardized root mean square residual (SRMR). The R^2 value for endogenous constructs exceeded 0.26, demonstrating substantial explanatory power. IC had a strong effect size ($f^2 = 2.14$) on OA but negligible effect on DTE ($f^2 = 0.01$), while OA had a medium effect size ($f^2 = 0.2$) on DTE. The Q^2 value of 0.29 indicated moderate predictive relevance for DTE, while OA had strong predictive relevance ($Q^2 = 0.56$). The SRMR value of 0.04 (<0.08) also further supported good model fit.

5.3. Mediation analysis

Mediation analysis tested the role of OA in the relationship between IC and DTE. Indirect, direct, and total effect were examined at both higher-order and lower-order levels using the bootstrapping routine.

Direct effects. Hypothesis H1 assessed IC's impact on OA and its sub-dimensions. IC significantly and positively influenced OA ($\beta = 0.83$, $t = 36.79$, $p < 0.001$). At the lower level, HC significantly and positively impacted WA ($\beta = 0.65$, $t = 16.63$, $p < 0.001$), SC influenced OpA ($\beta = 0.73$, $t = 21.32$, $p < 0.001$), and RC impacted NA ($\beta = 0.71$, $t = 22.37$, $p < 0.001$). The *p* values of all hypotheses were clearly below 0.01, suggesting that all hypotheses were significant at a level of 1 %. Hence H1, H1a, H1b, H1c were supported.

Hypothesis H2. evaluated OA and its sub-dimensions' effect on DTE. The results revealed that OA has a significant and positive impact on DTE ($\beta = 0.58$, $t = 7.41$, $p < 0.001$). At the lower level, WA ($\beta = 0.19$, $t = 2.51$, $p < 0.05$), OpA ($\beta = 0.22$, $t = 2.2$, $p < 0.05$), and NA ($\beta = 0.23$, $t = 2.42$, $p < 0.05$) positively impacted DTE. The p value of H2 was below 0.01, indicating that it was significant at a level of 1 %. Therefore, H2, H2a, H2b, H2c were supported. The results of direct relationships were presented in Table 2.

Indirect effects. Hypothesis H3 evaluates the mediating effect of OA (and its sub-dimensions). The results showed a significant indirect effect of IC on DTE through OA ($\beta = 0.48$, $t = 7.32$, $p < 0.001$), while the direct effect of IC on DTE with OA included was insignificant ($\beta = 0.13$, $t = 1.51$, $p = 0.13$). However, the total effect of IC on DTE was significant ($\beta = 0.61$, $t = 14.74$, $p < 0.001$), confirmng that OA fully mediated the relationship between IC and DTE; hence, H3 was supported and a full mediation (Baron and Kenny, 1986), also referred to as indirect-only mediation in Zhao et al.'s (2010) terminology, was established.

For sub-dimensions, HC had a significant indirect effect on DTE through WA ($\beta = 0.13$, $t = 2.46$, $p < 0.05$), but insignificant direct effect ($\beta = 0.02$, $t = 0.22$, $p = 0.83$) and total effect ($\beta = 0.14$, $t = 1.68$, $p = 0.09$), confirming WA as a full mediator (H3a). SC also showed a significant indirect effect on DTE through OpA ($\beta = 0.16$, $t = 2.2$, $p < 0.05$), while the direct effect was insignificant ($\beta = 0.11$, $t = 1.32$, $p = 0.19$), confirming OpA as a full mediator (H3b). Similarly, RC had a significant indirect effect on DTE through NA ($\beta = 0.16$, $t = 2.4$, $p < 0.05$), while direct effect ($\beta = 0.01$, $t = 0.19$, $p = 0.85$) and total effect ($\beta = 0.17$, $t = 1.89$, $p = 0.06$) were insignificant, supporting NA as a full mediator

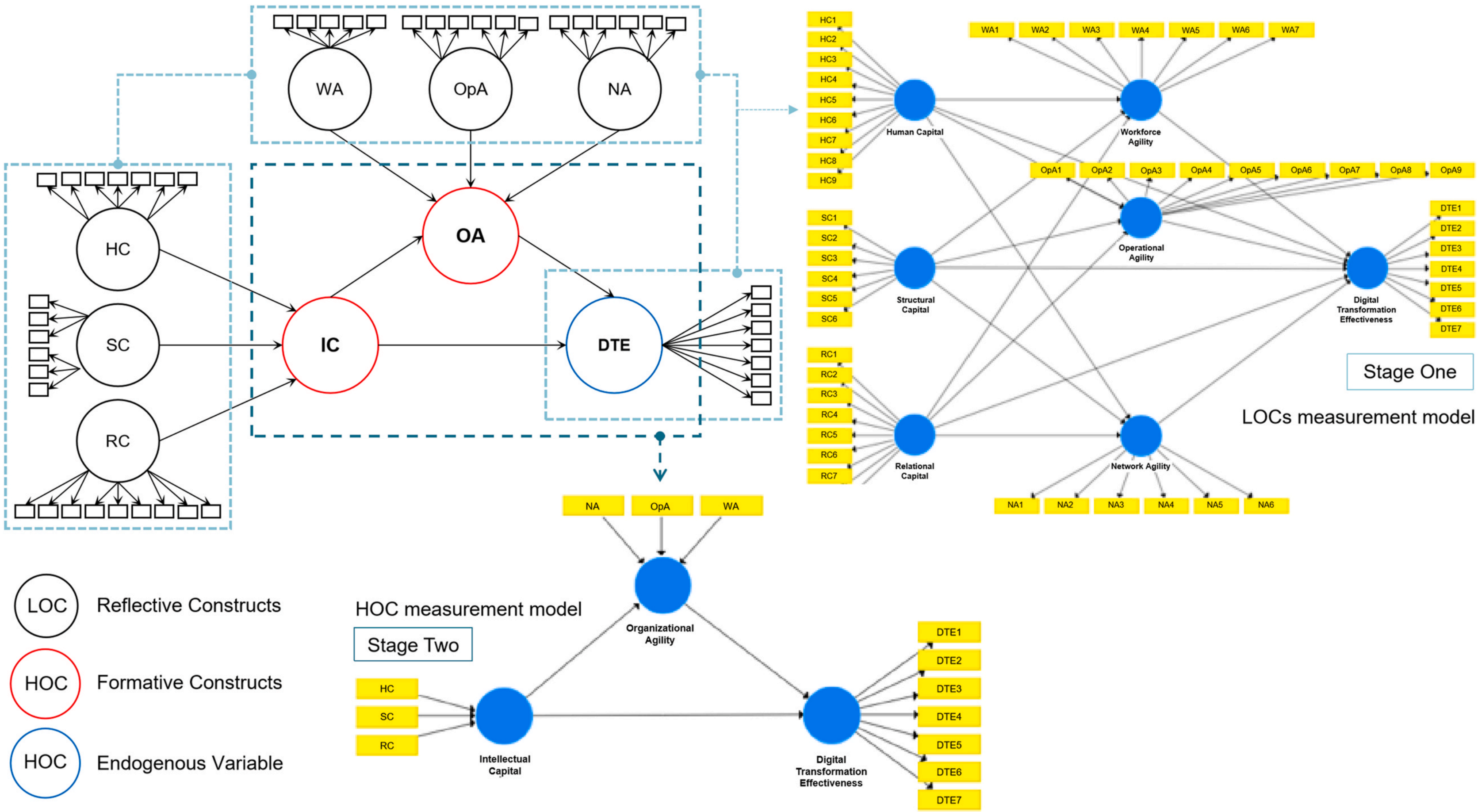

Fig. 2. Hierarchical component model overview using disjoint two-stage approach.

Table 2
The results of direct relationships.

Hypotheses	β	STDEV	T Values	P Values	Results
H1: IC - > OA	0.83***	0.02	36.79	0.00	Supported
H1a: HC - > WA	0.65***	0.04	16.63	0.00	Supported
H1b: SC - > OpA	0.73***	0.08	1.32	0.00	Supported
H1c: RC - > NA	0.71***	0.03	22.37	0.00	Supported
H2: OA - > DTE	0.58***	0.08	7.41	0.00	Supported
H2a: WA - > DTE	0.19*	0.08	2.51	0.01	Supported
H2b: OpA - > DTE	0.22*	0.1	2.2	0.03	Supported
H2c: NA - > DTE	0.23*	0.09	2.42	0.02	Supported

Note. STDEV = Standard Deviation. The significant ($p < 0.05$) path coefficients (β) are colored green. *$p < 0.05$; ***$p < 0.001$.

(H3c). Results are detailed in Table 3 and visualized in Fig. 3.

6. Discussion

Our findings confirm that IC alone is insufficient for DT success unless accompanied by a high degree of OA. This conclusion is supported by the empirical validation of our proposed model, which demonstrates that OA significantly mediates the relationship between IC and DTE. Specifically, the results show that while all three dimensions of IC—human capital, structural capital, and relational capital—have positive associations with OA, their direct effects on DTE become insignificant when OA is accounted for in the model. The full mediation observed in our structural equation modeling provides strong evidence that OA is the key enabling mechanism through which knowledge resources are translated into adaptive digital transformation outcomes. OA processes the organization's adaptive responses, breaking down hierarchical silos and fostering cross-functional collaboration and innovation, thereby transforming IC into catalyst for effective DT outcomes. This reinforces the central argument that IC must be dynamically mobilized through agile capabilities to deliver meaningful transformative results. This study underscores the necessity of OA as a dynamic capability that enables organizations to capitalize on IC investments for DT. This aligns with the finding that investing in continuous and sustained IC is of great importance to achieve radical innovation results in DT process (Delgado-Verde et al., 2016). Organizations must not only strengthen their IC but also develop OA to better support knowledge integration, resource reallocation, and agile decision-making. These findings align with previous studies (e.g., Gong and Ribiere, 2025; Warner and Wäger, 2019) that emphasize OA's role as a strategic renewal mechanism for successful DT.

6.1. Theoretical implications

This research contributes to DT, IC, and OA literature by offering a

Table 3
Mediation analysis results.

Hypothesis	Path	Effect	β	SE	T Value	P Value	Results
H3: IC - > OA - > DTE		Indirect	0.48***	0.07	7.32	0.00	Supported
	IC - > DTE	Direct	0.13	0.08	1.51	0.13	Rejected
		Total	0.61***	0.04	14.74	0.00	Supported
H3a: HC - > WA - > DTE		Indirect	0.13*	0.05	2.46	0.01	Supported
	HC - > DTE	Direct	0.02	0.08	0.22	0.83	Rejected
		Total	0.14	0.09	1.68	0.09	Rejected
H3b: SC - > OpA - > DTE		Indirect	0.16*	0.07	2.2	0.03	Supported
	SC - > DTE	Direct	0.11	0.08	1.32	0.19	Rejected
		Total	0.27**	0.1	2.79	0.005	Supported
H3c: RC - > NA - > DTE		Indirect	0.16*	0.07	2.4	0.02	Supported
	RC - > DTE	Direct	0.01	0.07	0.19	0.85	Rejected
		Total	0.17	0.09	1.89	0.06	Rejected

Note. STDEV = Standard Deviation. The significant ($p < 0.05$) path coefficients (β) are colored green. The insignificant ones are colored red. *$p < 0.05$; **$p < 0.01$: ***$p < 0.001$.

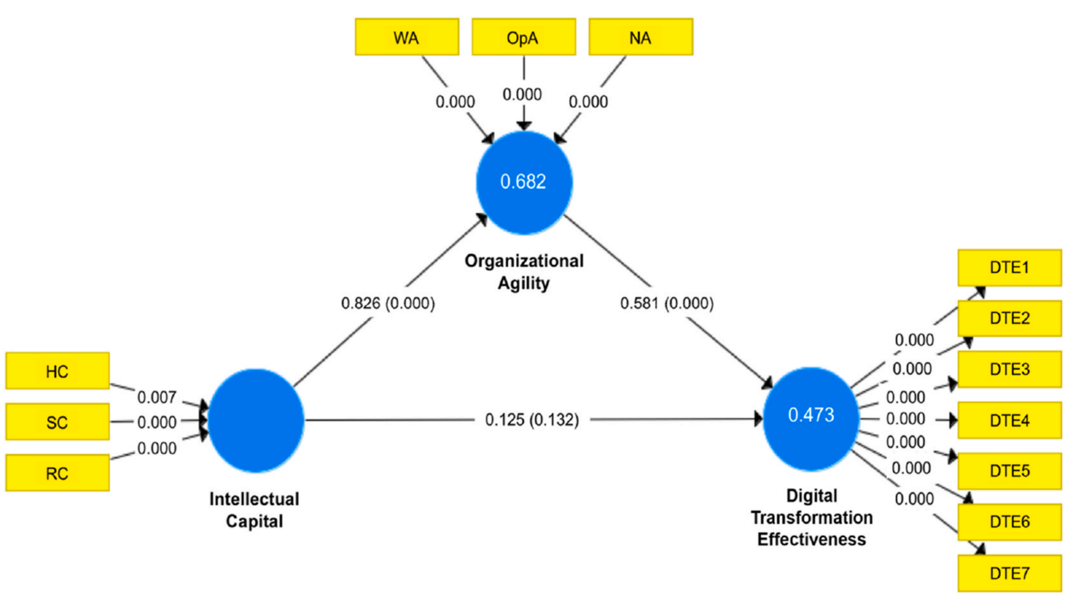

Fig. 3. Conceptual model with path coefficients, P-values, and R square.

novel perspective on how these constructs interact to drive DT success. Grounded in IC and dynamic capabilities theory, we extend prior studies by empirical testing OA's mediating role, providing theoretical clarity on the conditions under which IC translates into DT outcomes. This study advances theoretical understanding through four main contributions.

First, our study advances IC theory by conceptualizing how workforce agility, operational agility, and network agility facilitate knowledge mobilization across human, structural, and relational capital, respectively. Workforce agility ensures that employees' skills and knowledge remain adaptable, enabling continuous learning and responsiveness to the changing demands of DT. Operational agility supports the flexible reconfiguration of internal structures and processes, allowing organizations to capture technological opportunities. Network agility enhances knowledge exchange with external stakeholders, fostering dynamic partnerships crucial for DT success. In this context, workforce agility, operational agility, and network agility can be considered three lower-order capabilities that support internal and external knowledge flow, which together facilitate the building of higher-order dynamic capabilities – OA.

Second, our findings contribute to dynamic capabilities theory by reinforcing OA's dual role in facilitating both proactive and reactive responses to changes. In volatile DT environments, organizations must dynamically reallocate resources, innovate processes, and leverage IC to sustain competitive advantage. By integrating three dimensions of OA (i.e., workforce agility, operational agility, network agility), we further establish OA as a higher-order dynamic capability that enables organizations to balance stability and flexibility in DT. Thus, ambidextrous management of resources and practices is vital for achieving OA and, consequently, the successful DT outcomes.

Third, our study provides empirical support for the argument that DT effectiveness depends on the interplay between intangible assets (IC) and dynamic capabilities (OA). By quantitatively establishing these relationships, we fill a critical gap in existing research, offering a foundation of future studies to further examine OA-driven digital strategies. The demonstrated full mediation of OA suggests that even organizations with substantial IC will struggle to achieve DT success without an agile organizational structure and mindset. Identifying OA as a mediator reveals the underlying mechanisms through which IC drives DT, advancing knowledge in these fields and providing a basis for further research.

Fourth, this study extends the understanding of the IC-DT duality by highlighting its potential contribution to organizational sustainability. Specifically, when OA is developed through the alignment of IC and DT, organizations not only enhance their responsiveness but also strengthen their long-term digital resilience and long-term adaptability. These sustained dynamic capabilities enables organizations to remain competitive while maintaining environmental and social responsibilities in dynamic contexts. As suggested by Broccardo et al. (2024) and Li et al. (2024), this duality supports a trajectory of continuous value creation that is both innovative and sustainable.

6.2. Practical implications

Our findings have practical implications for business leaders and practitioner seeking to optimize DT success. From an intellectual capital perspective, organizations must treat people as essential capital rather than resources, actively managing and nurturing human, structural, and relational capital in alignment with DT goals. Organizations must strategically invest in developing their human capital (e.g., training, upskilling and reskilling for digital skills), structural capital (e.g., updating knowledge systems, digital infrastructures), and relational capital (e.g., building external innovation networks) as foundational pillars for digital success. Yet, organizations must recognize that intellectual capital alone is insufficient to drive effective DT; instead, it must be actively leveraged through OA-oriented management practices. For example, enhancing OA (e.g., fast decision-making, cross-functional teams, flexible structures, rapid experimentation hubs, adaptive cultures) is not optional — it is the mechanism that converts intellectual resources into real DT outcomes. This research offers four key practical implications for organizations.

First, senior executives play a pivotal role by setting strategic visions, fostering a culture of innovation, leveraging relational capital through strategic partnerships and collaboration for DT. They should dismantle rigid hierarchical structures, encourage cross-functional collaboration, and leverage relational capital through strategic partnerships. Establishing effective Objectives and Key Results (OKRs) can empower motivated, self-governed teams capable of responding dynamically to DT. Middle managers enhance OA by translating strategic goals into actionable plans, optimizing structural capital, and fostering continuous innovation. They drive operational agility by implementing flexible workflows, reallocating resources as needed, and institutionalizing knowledge-sharing mechanisms. Agile project management and iterative learning loops are essential to maintaining adaptability. Project team members directly contribute to workforce agility by continuously updating their skill setss, sharing knowledge, and proactively engaging in creative, experimental activities (e.g., innovation workshops and hackathons), and internalizing the local team's norms (Annosi et al., 2024). In a nutshell, senior executives focus on strategic foresight and external relationships, middle managers on operational processes and cross-functional coordination, and project team members on direct execution and innovation.

Second, bridging operational silos is critical for knowledge institutionalization and digital innovation, particularly when cross-functional collaboration is essential for achieving DTE. Effective knowledge management practices ensure seamless knowledge flow across employees, teams, and departments. This setup allows for the integration of OA into ongoing operations, promoting continuous learning and innovation, and laying the groundwork for smoother transitions between different operational initiatives. Organizations should develop flexible infrastructures to reallocate key resources (e.g., high-performing individuals, leadership time, expertise, required funding, infrastructures, and other key assets) swiftly as priorities shift or team performance varies. Organizations with well-developed IC are better positioned to leverage OA for higher DTE. Practitioners should prioritize enhancing OA to maximize IC's value and adopt agile practices that foster collaboration and adaptability.

Third, this research provides practitioners with a tool for assessing DT readiness by offering a rigorously tested, valid, and reliable survey instrument for organizational diagnosis. Organizations can use this diagnostic tool to identify their IC and OA strengths and weaknesses, enabling them to craft strategies for creating effective DT action plans and monitoring IC-based and OA-based metrics. By aligning IC development with OA-enhancing practices, organizations can build resilient and adaptable organizations capable of thriving in the digital age. In summary, our study offers a roadmap for optimizing DT outcomes by integrating IC and OA. Siloed knowledge and rigid structures inhibit both OA and DT. Thus, we emphasize that fostering a knowledge-centric culture and agile practices is critical for navigating DT complexities.

Fourth, our findings underscore that DT should be reframed as a capability-development journey rather than a one-time technology upgrade. When driven by strategically deployed knowledge assets—such as employee expertise, innovative processes, and relational trust—digital initiatives can support not only operational efficiency but also broader sustainability goals. Managers are encouraged to embed sustainability-oriented principles into the design of DT initiatives and to develop IC in ways that prioritize green innovation, ethical data use, and inclusive knowledge sharing. This aligns with Riaz et al. (2024), who advocate integrating green IC and green IS practices to support sustainable organizational performance.

6.3. Limitations and future avenue

This research faced three main limitations due to its methodology and design: limited internal validity, potential of social desirability bias, and a small sample size. Using a non-experimental correlational design, we measured the proposed variables without intervention, limiting conclusions to relationships rather than causal connections, thus affecting internal validity. Self-reported responses, though anonymized, may be influenced by social desirability bias. Efforts to minimize this bias included clear instruction, video explanations, and options to skip personal questions or withdraw from the survey at any time. The sample size of 265 met minimum requirements using the inverse square root method but may limit result generalizability. However, as this study is exploratory in nature and use a multi-stage sampling design, the impact of these limitations is mitigated.

Future studies should address these limitations through experimental design with random assignment, longitudinal analyses, or additional control mechanisms to better determine causal relationships. This study focuses on effectiveness from a performance-centric view and does not explicitly measure environmental or social sustainability. Future research could explore the sustainability-performance linkage of DT initiatives. A longitudinal approach could clarify how proposed constructs evolve over time and may help uncover how the development of dynamic capabilities in DT contributes to sustainable competitiveness. Future research could explore whether OA fits best in high-velocity markets and assess workforce agility, operational agility, and network agility in medium-velocity environments. Additionally, examining whether environmental dynamism and sustainability orientation moderates mediated relationships or impacts full mediation would be valuable. Moreover, including more objective indicators (e.g., Number of new digital offerings launched, Revenue growth from digital streams, Time-to-market for new initiatives, etc.) alongside perceptual measures could enhance predictive validity, offering deeper insights into the proposed model.

7. Conclusions

In an era marked by heightened uncertainty, globalization, and accelerated technological change, digital transformation has become both a necessity and a challenge for organizations seeking sustained competitiveness. Amid these complexities, this study advances our understanding of how organizations can effectively navigate digital transformation by leveraging their intangible assets—specifically, intellectual capital—through the dynamic capability of organizational agility. This study provides novel insights into the mediating role of organizational agility, highlighting how its distinct dimensions (i.e., workforce agility, operational agility, and network agility) serve as critical conduits through which intellectual capital is translated into digital transformation effectiveness. This research offers a comprehensive and empirically supported framework that enables organizations to better assess, structure, and enhance their digital transformation initiatives. It delivers strategic and actionable insights for key organizational stakeholders, including senior executives, middle managers, and project teams. From a theoretical perspective, this research contributes to the knowledge and innovation management literature by illuminating the mechanisms through which intellectual capital drives transformation outcomes in fast-changing environments. It underscores that possessing intellectual capital alone is insufficient; instead, organizations must also cultivate organizational agility to mobilize and apply that knowledge strategically to drive innovation. To remain adaptive and forward-looking in today's volatile business landscape, our empirical results demonstrated that organizations should invest in their capability to balance both their efforts to pursue new and emerging opportunities and their capabilities to take advantage of existing resources and assets. Our results underscores the importance of developing both the reactive and proactive facets of organizational agility to achieve digital transformation. The findings on the significant multifaceted mediation of organizational agility revealed that the timely and effective deployment of intellectual capital's sub-dimensions is essential to maximizing digital transformation effectiveness. In sum, this study reinforces the imperative for organizations to embed organizational agility into their digital strategies and underscores the value of a dynamic, knowledge-driven approach to transformation success.

CRediT authorship contribution statement

Cheng Gong: Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Vincent Ribiere:** Writing – review & editing, Validation, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

The Summary of Survey Questions and Their Sources.

	Items		Sources
Human Capital (HC)	1	Employees in our organization are highly skilled in their respective jobs.	Zhang et al. (2018), Subramaniam and Youndt (2005)
	2	Employees in our organization have very up-to-date knowledge and capabilities, relative to our direct competitors.	Claus (2019)
	3	Employees in our organization are experienced.	Han and Li (2015), Subramaniam and Youndt (2005)
	4	Employees in our organization usually come up with novel and valuable ideas.	Sharabati et al. (2010), Bontis (1998)
	5	Employees in our organization are keen to voice their opinions and ideas in group discussions.	
	6	Employees in our organization regularly find creative solutions for problem-solving.	
	7	Our organization's employees undergo continuous training programs every year.	Sharabati et al. (2010), Bontis (1998)

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(*continued*)

		Items	Sources
Structural Capital (SC)	1	Our organization incorporates much of its experiential knowledge in its processes and practices.	Engelman et al. (2017), Ferenhof et al. (2015), Carmona-Lavado et al. (2010), Youndt et al. (2004)
	2	Our organization supports the internal dissemination of information and the flow of knowledge.	
	3	Our organization's culture and atmosphere are supportive to knowledge sharing.	Sharabati et al. (2010), Bontis (1998)
	4	Our organization actively involves employees in idea generation and the implementation of innovations.	De Jong and Vermeulen (2006)
	5	Our organization has a team/unit which is continuously seeking for and providing support to innovative opportunities.	
	6	Our organization has innovation governance mechanisms (e.g., innovation committees, task forces or teams, etc.) to align goals, allocate resources and assign decision-making authority for innovation.	Deschamps (2012)
Relation Capital (RC)	1	Our major external stakeholders and us keep each other informed about events or changes that may affect the other party.	Sambasivan et al. (2011, 2013)
	2	Our major external stakeholders communicate well their expectations for our performance.	Morgan and Hunt (1994)
	3	The relationship that we have with our major external stakeholders is expected to continue on a long-term basis.	Villena et al. (2011)
	4	Our organization capitalizes on customers' wants and needs by continually striving to make them satisfied.	Sharabati et al. (2010), Bontis (1998).
	5	Our customers' knowledge is widely distributed throughout our organization.	
	6	Our organization is keen on developing long-term relationship with our customers.	AI-Jinini et al. (2019), Engelman et al. (2017)
	7	Our organization currently has many active joint projects with our diverse partners (e.g. R&D, manufacturing, marketing, distribution channels).	Sharabati et al. (2010), Bontis (1998)
	8	Our organization has developed tight and trusted relationships with our partners.	
	9	Our organization maintains a long-standing relationship with our partners.	Sharabati et al. (2010), Bontis (1998)
Workforce Agility (WA)	1	We are able to identify and seize rapidly the best opportunities which come up in our environment.	Charbonnier-Voirin (2011)
	2	We foresee a wide range of scenarios and prepare future plans accordingly.	Doz and Kosonen (2010), Yang and Liu (2012)
	3	We regularly conduct local experiments to test and learn from the markets.	Doz and Kosonen (2010)
	4	We have a good understanding of how our own job relates to the organization's overall objectives.	Charbonnier-Voirin (2011), Hoyt et al. (2007)
	5	We are willing to continuously learn from one another and to pass useful and important knowledge to others (within and cross different business units).	Misra et al. (2009)
	6	Our organization's processes enable us to make decisions quickly when circumstances change.	Charbonnier-Voirin (2011)
	7	Our employees have a lot of autonomy in their work.	
Operational Agility (OpA)	1	We can reorganize in flexible and empowered team configurations to meet changing requirements and new challenges.	Misra et al. (2009)
	2	Our organizational structure allows for flexible redeployment of our resources.	Hock et al. (2016), Doz and Kosonen (2008, 2010)
	3	Our organization's underlying business systems and processes are modular and can be easily changed.	Ogunleye et al. (2021)
		Our organization has information systems and technologies that …	
	4	… are integrated amongst different departments and/or business units.	Vázquez-Bustelo et al. (2007)
	5	… provide information helping our employees to quickly respond to changes.	Zelbst et al. (2011), Kassim and Zain (2004)
	6	… enable us to fully integrate our customers and partners into our processes.	Vázquez-Bustelo et al. (2007)
	7	We fulfill demands for rapid-response, special requests of our customers whenever such demands arise.	Felipe et al. (2017, 2020), Lu and Ramamurthy (2011)
	8	We can quickly scale up or scale down our production/service levels to support fluctuations in demand from the market.	
	9	We can quickly make necessary alternative arrangements and internal adjustments whenever there is a disruption in supply from our suppliers.	
Network Agility (NA)	1	We are quick to make and implement appropriate decisions in the face of market/customer-changes.	Felipe et al. (2017), Lu and Ramamurthy (2011)
	2	We constantly look for ways to reinvent/reengineer our organization to better serve our market place.	
	3	We closely collaborate with and encourage fast feedback from our customers.	Misra et al. (2009)
	4	We are able to quickly exploit the resources and capabilities of partners to enhance the quality and quantity of products and services.	Felipe et al. (2017), Yang and Liu (2012),
	5	We are able to dynamically manage relationships with outsourcing partners.	Bradley et al. (2012),
	6	We can easily switch partners to avail ourselves of lower costs, better quality or improved delivery times.	Tallon and Pinsonneault (2011)
Digital Transformation Effectiveness (DTE)	1	Imagine an ideal organization utilizing digital technologies and capabilities to improve processes, engage talent across the organization, and drive new and value-generating business models. How close is your organization to that ideal?	Kane (2019)
	2	In general, the DT projects in our organization are well-defined with clear goals and objectives.	
	3	In general, the DT projects in our organization achieve emerging objectives in the implementation processes.	
	4	In general, the DT projects in our organization meet the internal stakeholders' satisfaction.	
	5	In general, the DT projects in our organization meet the external stakeholders' satisfaction.	
	6	In general, the DT projects in our organization re-innovate business model.	
	7	In general, how would you characterize the outcome of digital transformation projects in your organization to date?	Kane (2019)

Data availability

The data that has been used is confidential.

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